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Abstract

Login page is a web-based platform that empowers developers to **build, preview, and showcase** web applications in a seamless environment. The platform features a **login/signup system**, **project showcase area**, **embedded analytics**, and **real-time preview functionality**.

Developed using **HTML, CSS, JavaScript**, and designed for future backend integration with **PHP and MySQL**, DemoBase delivers:

- A modern landing page with a responsive layout
- Features section explaining core offerings
- About and contact sections
- A dynamic UI with Flexbox layout
- Embedded map and styled form for contact functionality

This project represents the frontend and structural foundation of a full-stack deployment platform for demo applications.

1.Introduction

This project implements a **Student Portal System** using PHP and MySQL, combined with HTML, CSS, and JavaScript for frontend interaction. The portal is designed to allow students to **register**, **log in**, and **access personalized content** securely. It demonstrates fundamental principles of **web development**, **session management**, **form validation**, and **secure data handling**.

2. Technologies Used

- Frontend: HTML5, CSS3 (responsive with media queries), JavaScript (for client-side form validation).
- Backend: PHP (server-side scripting), MySQL (relational database).
- Security: Password hashing using password_hash() and verification using password_verify().
- Database Connectivity: MySQLi with prepared statements to prevent SQL injection.

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3. Database Design

Database: logindbs
Table: Signin

Structure of the Signin table:

Column Name	Type	Description
Name	VARCHAR(255)	Student's full name
Age	INT	Student's age
Mobile_no	VARCHAR(10)	Unique 10-digit mobile number
Password	TEXT	Encrypted password (hash)

The Mobile_no acts as a unique identifier for each student.

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4. Functionalities Implemented

Signup (Registration)

- Located under ?page=signup
- Collects:
  - Name
  - Age
  - Mobile number
  - Password
- Client-side validation:
  - Mobile number: exactly 10 digits

    - Age: between 1 and 120
- Server-side validation:
    - Checks for existing mobile number
    - Verifies age and mobile formats
- Security:
    - Password is hashed before storing in the database using password_hash() with the PASSWORD_BCRYPT algorithm.

Login

- Located under ?page=login
- Collects:
    - Mobile number
    - Password
- Authenticates user by:
    - Matching the mobile number in the database
    - Verifying the hashed password using password_verify()
- On successful login:
    - Stores session variable $_SESSION['user']
    - Redirects user to ?page=about
    - Displays a welcome message with auto-redirect and logout option

Session Management

- Uses session_start() to manage user sessions.
- On login, session variables store user data for personalized experience.
- On logout (?page=logout):
    1. session_unset() and session_destroy() are called.

2. Redirects user to the login page.

Dashboard / About Page

- Accessible only if $_SESSION['user'] is set.
- Displays:

1. A personalized welcome message
2. Essay about the portal's purpose and technologies used
3. Code snippet for database connection
4. Images related to student life (from Unsplash)

Logout

- Clicking the Logout button triggers a form that sets ?page=logout.
- Ends the session and redirects to login.

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5. Code Structure

- Page routing:
  - Controlled using $_GET['page'] (e.g., ?page=signup, ?page=login, etc.)
  - login, signup, and about are handled conditionally in the PHP code.
- **Error Handling:**
  - Signup and login errors are stored in variables ($signup_error, $login_error) and shown in HTML.
- **Password Handling:**
  - On signup: password_hash()
  - On login: password_verify()
- **SQL Injection Prevention:**

- All database queries are executed using prepared statements with bind_param().

- **CSS Styling:**
  - Includes a modern UI using gradients, shadows, and responsive design.
  - Flexbox is used for layout alignment.
  - Includes hover effects and mobile-friendly adjustments.
- **JavaScript:**
  - Used only for signup validation (age range and mobile number).

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6. Security Considerations

- Passwords are never stored in plaintext.
- Sessions are used instead of persistent cookies.
- SQL injection is mitigated via prepared statements.
- Suggestions for Improvement:
  1. Add CSRF tokens in forms to prevent cross-site request forgery.
  2. Set session cookie flags like HttpOnly and Secure.
  3. Implement email verification or OTP (optional).

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7. User Experience (UX)

- Responsive design for all devices (mobile, tablet, desktop).
- Clear error messages and success feedback.
- Uses meta refresh for timed redirection post-login.
- Smooth navigation between login and signup pages.

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8. Future Improvements

- Profile page with editable student details
- Forgot password functionality
- Admin dashboard to manage users
- Email-based authentication or OTP
- Notification system for important announcements

Conclusion

Website management is a crucial aspect of maintaining an effective and professional online presence. It involves a combination of technical, design, and strategic tasks that ensure the website remains functional, secure, up-to-date, and user-friendly.

A well-managed website enhances user experience, improves search engine rankings, and supports business goals such as customer engagement, lead generation, and online sales. Key components of website management include:

- **Regular content updates** to keep information relevant.
- **Security monitoring** to protect against threats.
- **Performance optimization** to ensure fast loading times.
- **Backup and recovery systems** to prevent data loss.
- **Analytics tracking** for data-driven decisions.